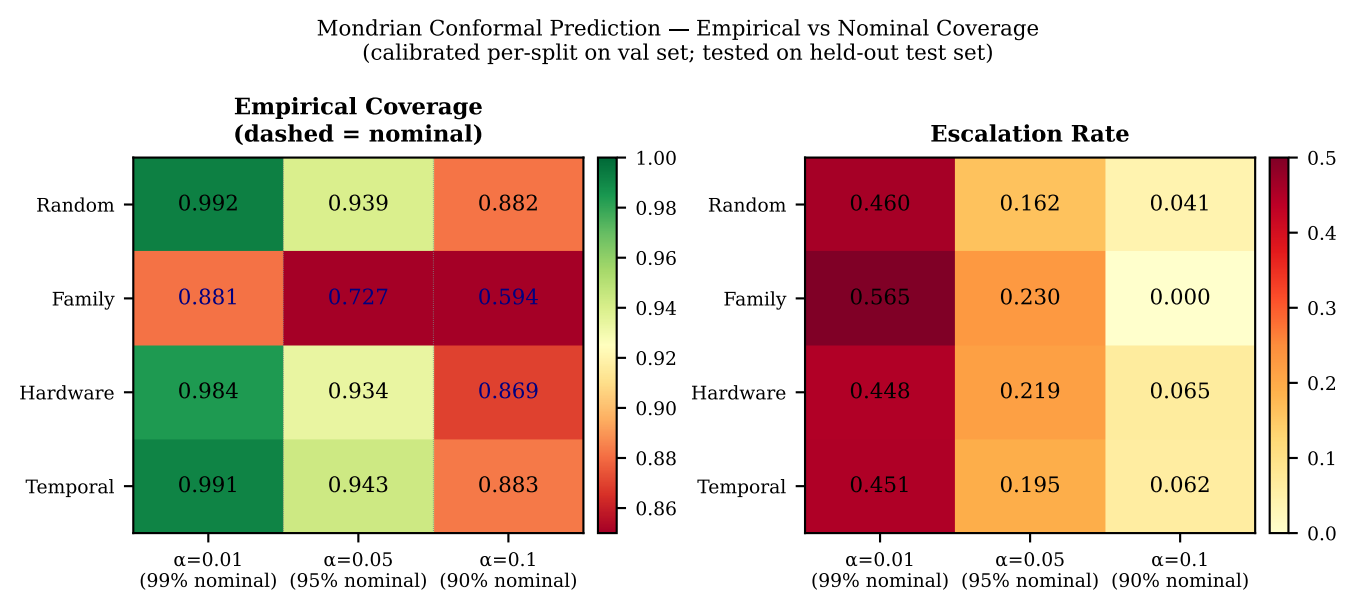
Mondrian Conformal Prediction — Empirical vs Nominal Coverage
(calibrated per-split on val set; tested on held-out test set)
Empirical Coverage
(dashed = nominal)
Random
Family
Hardware
Temporal
0.992
0.939
0.882
0.881
0.727
0.594
0.984
0.934
0.869
0.991
0.943
0.883
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)
Escalation Rate
Random
Family
Hardware
Temporal
0.460
0.162
0.041
0.565
0.230
0.000
0.448
0.219
0.065
0.451
0.195
0.062
0.5
0.4
0.3
0.2
0.1
0.0
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)